1. (12 pts)

4PTS (a) Find the area of the triangle through the points $A(1,2,1)$, $B(3,1,2)$, and $C(2,4,3)$.

Area = ______________________

4PTS (b) Consider the curve $y = -x^2 + 4x + \sin(x-1)$. At the point where $x = 1$, give a unit vector that is parallel to the tangent line and a unit vector perpendicular to the tangent line.

a unit vector parallel to tangent = ______________________

a unit vector perpendicular to tangent = ______________________

4PTS (c) Find the center and radius of the sphere with P such that the distance from P to $(0,0,4)$ is triple the distance from P to $(0,0,0)$.

Center = ______________________

Radius = ______________________

2. (14 pts)

2 PTS (a) Assume **a**, **b** and **c** are vector in 3-dimensions. For each of the following expressions write if the result would be 'a Vector', 'a Scalar', or 'Meaningless'.

+1 i. $(\mathbf{a}+\mathbf{b})\cdot\mathbf{c}$ is ______________________

+1 ii. $\mathbf{a}\times\mathbf{b}-\mathbf{c}$ is ______________________

6 PTS (b) Find parametric equations for the line of intersection of the planes $x+2y-z=4$ and $2x-y+z=1$.

Line Equation: ______________________

6 PTS (c) Find the equation of the plane that passes through the point $(1,2,3)$ and contains the line $x=2+t$, $y=1+t$, $z=4+2t$. Then find the z-intercept of this plane.

Plane Equation: ______________________

z-intercept: $(x,y,z)=$ ______________________

3. (12 pts)

2PTS (a) Give the precise name of the surface given by the equation $4 = x^2 - 2y^2 - z^2$.

Surface name: ______________________________

4PTS (b) Find parametric equations for the curve of intersection of the cylinder $x^2 + y^2 = 4$ and the plane $z = x - y$. Then set up, but do **not** evaluate, an integral for the arc length of this curve for one complete trip around the cylinder.

Curve Equations: ______________________________

Length Set-Up: ______________________________

6PTS (c) The curves $\mathbf{r}_1(t) = \langle 2t+1,\ 6-t,\ t^2+2t+1 \rangle$ and $\mathbf{r}_2(u) = \left\langle u-2,\ u,\ \frac{u+3}{2} \right\rangle$ intersect at a point. Find the point of intersection and the angle between the curves at that point. Give your final answer for the angle either in simplified exact form or as a decimal rounded to the nearest degree.

Intersection Point: $(x, y, z) =$ ______________________________

Angle = ______________________________ degrees

4. (12 pts)

(a) Consider the curve $\mathbf{r}_1(t) = \langle t^2 + 1,\ 3t - 2,\ t^3 - t \rangle$.

4pts

i. Find parametric equations for the tangent line to the curve at the point $(2, 1, 0)$.

Tangent Line Equations: __________ , __________ , __________

4pts

ii. Find the point where this tangent line intersects the xz-plane.

Intersection point $(x, y, z) =$ ____________________

4pts

(b) Find the curvature of the curve $\mathbf{r}_2(t) = \langle t,\ t^2,\ 2t \rangle$ at $t = 1$. Hint: Remember to plug in $t = 1$ as early as possible. You can leave your final answer in unsimplified exact form or as a decimal rounded to 3 digits.

$\kappa(1) =$ ____________________